

IRB PROTOCOL

Title:	A Graph-Augmented RAG-based Tool for Automated Bug Localization and Repair Guidance
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Faculty Advisor/ Principal Investigator:	Kazi Muheymin-Us-Sakib

Before starting your protocol submission, please make sure you have reviewed the following resources:

1. [DU IRB Guidance Page](#)
2. [Research Integrity Framework](#)

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Getting Started

1. Which committee will review the submission? (*Highlight one.*)

- ☐ StudentIRB
- ☐ PsychIRB
- ☐ EdIRB
- ☒ IRB

2. What [category](#) will your research fall into? (*Highlight one.*)

- ☒ Exempt
 - Category #: Category 2: Educational tests, surveys, interviews, or observation of public behavior
- ☐ Expedited
 - Category #:
- ☐ Full Committee

Study Personnel

List the personnel who will be involved in your project, along with their role. Add as many rows as needed.

Name	Email	Role
Jitesh Sureka	bsse1115@iit.du.ac.bd	Co-Principal Investigator (Primary Contact)
Kazi Muheymin-Us-Sakib	sakib@iit.du.ac.bd	Principal Investigator

Study Design

1. Study Background, Hypothesis, and Objectives

Bug localization is a fundamental activity in software maintenance. It is the task of identifying the specific files, classes, or functions responsible for a reported defect [9]. It directly affects how quickly developers can understand and resolve issues, particularly in large and complex codebases. Relying solely on manual

bug localization can be time-consuming and error-prone [32]. This motivates the development of automated approaches that can efficiently search large codebases and retrieve the most relevant code elements linked to a reported bug.

However, existing tools still struggle to exploit full repository context, such as dependencies, configuration, and error-handling logic regarding where a bug lies or how to repair it. Existing tools often struggle to offer repository-level bug localization that makes use of the full context of an entire codebase.

To address this, we propose Insight, a repository-aware tool that combines semantic code search, structural graphs, and LLM reasoning to rank locations and generate explanations and repair suggestions. To evaluate the usability and usefulness of INSIGHT, we aim to conduct a user study survey. This study will evaluate Insight with developers. We hypothesize that Insight will correctly localize most bugs at the file level and that its explanations will be perceived as useful for debugging. Our objectives are to measure localization performance and to gather feedback on how Insight’s explanations influence developers’ workflows.

2. Study Method and Procedure

We will conduct a user study with 30-35 professional developers and academics, each with two to ten years of experience working with medium-to-large codebases. Because bug localization and repair are common across both industry and academia, we will focus on participants who regularly interact with issue trackers and debugging workflows. Our goal is to evaluate Insight’s (i) usability and perceived usefulness (RQ1/RQ2), (ii) quality of its explanations (RQ3), and (iii) effectiveness in capturing repository context for localization and remedy suggestion (RQ4).

We will randomly select one repository [18] from the Long Code Arena dataset [5] and clone it into a separate environment. From this repository, we will select four real issues: two where Insight is expected to accurately localize bugs and two where its predictions are weaker. Participants will receive a short written tutorial on Insight and will be asked to work with the cloned repository, examine the four issues, run Insight, and review the ranked candidate locations and explanations.

After this interaction, participants will complete a survey combining 5-point Likert items with open-ended questions on usability, usefulness, explanation quality, and contextual awareness. We will analyze

closed-ended responses using descriptive statistics and simple comparative checks between “successful” and “less successful” issue cases. Open-ended responses will be coded independently by two researchers and reconciled into shared themes. This analysis is expected to provide a nuanced view of Insight’s strengths and limitations in realistic debugging workflows and to inform our design recommendations for repository-aware bug localization tools.

3. Will you be collecting or accessing any of the following? (*Highlight all that apply.*)

- ☐ Audio
- ☐ Video
- ☐ Medical Records
- ☐ Educational Records
- ☐ Urine
- ☐ Saliva
- ☐ Blood
- ☐ Other:

4. Data Analysis Plan

We will use descriptive statistics (means, medians, standard deviations, and response distributions) for Likert-scale items and, when appropriate, nonparametric tests to explore differences between issue conditions. Open-ended survey responses will undergo multi-coder inductive thematic analysis.

5. Study Dates

- a. Study Start: 1st January, 2026
- b. Study End: 12th January, 2026

6. External Sites

Since this study can be conducted remotely using a personal laptop, we will be able to use a wide range of participants from different geographic locations, across different software organizations, and academic institutions.

Participant Recruitment

1. Recruitment Process

The target participants for this study will be individuals involved in the development or maintenance of medium-to-large software systems, particularly those with hands-on experience in debugging, bug localization, and using issue trackers (e.g., GitHub Issues). All participants will be 18 years or older, with no restrictions on gender or nationality.

We will identify potential participants through professional and personal networks, departmental and lab mailing lists, and online communities (e.g., LinkedIn, university Slack/Discord channels). Recruitment will be conducted via email and brief online announcements describing the study purpose, eligibility criteria, approximate time commitment, and the voluntary nature of participation. Interested individuals will be directed to a study information and consent page before enrollment. Copies of recruitment messages and announcement text will be provided as supporting documents in the IRB submission.

2. Ages

18 and older

3. Vulnerable Populations

Our study will not include vulnerable populations.
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4. Screening Process and Exclusion Criteria

The only selection criteria for a participant will be familiarity with GitHub repository management, issue management, and debugging and developing software projects. We will not screen the user's abilities, because we want a variety of participants working on different kinds of software projects, handling different types of bugs every day.
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5. Participant Enrollment and Locations

- Total Expected Enrollment:

35, but it could be more than this, if needed

- Study Locations (i.e., where will the data be collected):

Remotely, on campus

6. Participant Duration

Each user will be able to complete the study virtually and without our intervention. The study consists of one survey, which will last approximately 25-30 minutes.

7. Participant Incentives

N/A

Participant Protection**1. Potential Risks to Participants**

It is possible that user responses could be accessed by an unauthorized user (lost device, compromised system, etc.). However, we will mitigate this risk by only storing anonymous data and by using best security practices to protect remotely stored data. Our data will not include personal information (age, name, organization name, project name, etc), and therefore, the data can not be traced back to a specific participant. We also encourage participants not to share any confidential organizational data or trade secrets.

2. Potential Benefits

There are potentially no direct benefits to the participants. They may, however, gain some personal insight into their own debugging and bug-localization practices by reflecting on their experiences and interacting with the Insight tool.

At a broader level, participants' responses will help us better understand how developers use repository-aware bug localization tools, what kinds of explanations and context they find useful, and where current tooling falls short. This can inform the design of future tools, workflows, and best practices for debugging, ultimately benefiting the wider software engineering community
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3. Participant Privacy & Confidentiality

Survey and interview completion are completely voluntary and completed digitally. Respondents may withdraw at any time. As previously mentioned, responses obtained from both the survey and the interview will be de-identified to protect respondents. The survey will not require any personal information from the participants, and all responses will be anonymized to protect the participants. The investigator's briefing to participants will include a section explicitly instructing participants not to share any personal information or any confidential organizational information in the survey. If a participant provides any personal information in the survey, investigators will exclude all personal information from data analysis. When the results of the study are published in the final paper, no personal identifiers will be included with any of the data displayed.

4. Risk and Safety Monitoring

Will your project pose more than *minimal* risk? -> **No**

5. Debriefing Script

After the survey, respondents will see the default Google Form end-of-survey screen that notifies them that their submission has been recorded.

6. Participant Complaints

Study participants will be able to reach out to all investigators through email if any complaints or questions arise.

7. Adverse Events Reporting

It is not likely that an adverse event (participant confidentiality breached, etc) will occur, but if it does, we will notify the IRB within 7 days of the incident via email. Together with the IRB committee, we will make a strategy to remedy the event.

8. Special Arrangements (Leave blank if n/a)

N/A

Data Collection and Security

1. Collection & Handling of Data

All information collected about the activity of the participants will be kept confidential. The information will be kept in digital format, which can be accessed only by the investigators (i.e., the proper lock and key conditions will be enforced).

Possibly identifiable information we will be tracking includes-

- Years of experience with software development, debugging, and issue management
- Past domains of experience
- Email address
- Type of the project the participant is working on

which will be optional to provide and serve only to facilitate a possible follow-up interview. These will not be included in our dataset. Identifiable information will be stored securely in a shared Box drive. When the results of this research are published or discussed at conferences, no information will be included that would reveal the identity of any of the participants.

2. Data Storage During Study: Data & Specimen Security

Researchers will maintain a secure database. All data will be stored electronically in that database. Survey responses with the Google Forms system will be directly connected to the database. All consent forms will be electronically stored as part of the survey in the database. When stored remotely, survey and interview responses (or derivatives containing PII) will be hosted securely using the Box file sharing system protected with two-factor authentication. Only investigators will have access to this data. Any data that is to be published in the final article or replication package will be appropriately de-identified to preserve respondent privacy.

3. Data/Specimen Banking & Future Research

The data will be stored in a secure location in one of the servers of the research lab, which is maintained by the IIT, University of Dhaka, and the lab. Survey responses may be used for a potential extension to this work. Only the listed investigators will be able to access the artifacts. Any identifiable data will be deleted upon publication of the results, while de-identified data will be stored indefinitely to assist with future research and will be made publicly available as part of a replication package.

4. Sharing Results

The participants will know about the results of the study when the final paper is published at a conference or journal venue. The contact information of the investigators will be shared with the participants, and the participants can contact the investigators about the results of the study following its release.

5. Identifiable Information to be collected

Highlight all identifiers or “none of the above” that will be recorded for this project.

- Patient/Participant Names
- Geographic subdivisions smaller than a State (with exceptions for the initial 3 digits of zip code)
- Age information for those over 89
- All elements of Dates (except year) that are directly related to an individual
- Telephone Numbers
- Fax Numbers
- Email Addresses
- Social Security Numbers
- Medical Record Numbers
- Account Numbers
- Health Plan Beneficiary Numbers
- Certificate/License Numbers
- Device Identifiers and Serial Numbers
- Vehicle Identifiers and Serial Numbers (including license plate numbers)
- Web Universal Resource Locators (URLs)
- Internet Protocol (IP) Address Numbers
- Biometric Identifiers (including finger and voice prints)
- Full face photographic images and any comparable identifying images
- Any other unique identifying number, characteristic, or code
- **None of the above**

6. Storage and Maintenance of Identifiers

All direct identifiers (e.g., names, email addresses, GitHub usernames, Zoom IDs) will be stored separately from research data in an encrypted file. For analysis and publication, we will use only de-identified, coded data. Within three years of project completion, the identifier file and any re-identification key will be permanently destroyed, leaving only anonymous datasets.

7. **Health Information Collected:** If you are NOT collecting health information, please skip this question.

a. **Sharing of personal health information (PHI):**

b. **Protection of PHI:**

Consent

1. **Describe the procedures for obtaining informed consent.**

Before a respondent can take our survey, they must read and agree to the consent form (see attached). The consent form is the first question of the survey and gates all further questions. The respondent will see the full consent text and then be asked to indicate consent by selecting either 'Yes' or 'No' from a radio button selection. Respondents are not able to move forward to the survey before this step. The consent information is stored along with the other survey information collected by Google Forms. The consent form shown on Google Forms may be slightly different from the one attached, due to limitations with Google Forms' display formatting.

2. **Non-English-Speaking Subjects** *[complete only if applicable]*

N/A

3. **Cognitively Impaired Adults/ Use of a Legally Authorized Representative (LAR)** *[complete only if applicable]*

N/A

4. **Consent Waivers**

a. Will your study require a waiver of consent?

N/A

b. Will you be requesting an alteration of consent or will you be altering or removing any of the required Elements of Consent? (***This is applicable if your study is using deception.***)

No

c. Will you be requesting a Waiver of Documentation of Consent? No

o **Highlight** one of the following

a. The only record linking the subject and the research would be the Informed Consent form and the principal risk would be potential harm resulting from a breach of confidentiality. Each subject will be asked whether the subject wants documentation linking the subject with the research, and the subject's wishes will govern.

b. The research presents no more than minimal risk of harm to subjects and involves no procedures for which written consent is normally required outside of the research context.

- c. If the subjects or legally authorized representatives are members of a distinct cultural group or community in which signing forms is not the norm, that the research presents no more than minimal risk of harm to subjects and provided there is an appropriate alternative mechanism for documenting that informed consent was obtained.
- How will the rights and welfare of participants be protected even though you will not be obtaining documentation of informed consent from the participant? Provide a rationale explaining why this Waiver of Documentation of Consent will not adversely affect the rights and welfare of the participants.

N/A

References

Include any references below in the appropriate format.